

conservation of energy - practice problems

name _____

1. A 2.0 kg cart rolls down a curved, frictionless track and drops 1.5 m in height. How much gravitational potential energy does it lose, and how much kinetic energy does it gain?

2. A skateboarder is moving at 6.0 m/s at the bottom of a smooth hill. Ignoring friction, how high up the hill can the skateboarder coast before stopping?

3. A 4.0 kg cart moving at 5.0 m/s rolls up a frictionless track. How much kinetic energy does it still have at a point 0.80 m higher?

4. A 2.0 kg cart starts from rest and rolls down a frictionless ramp that is 5.0 m long. How fast is it moving at the bottom?

5. A 0.30 kg cart is launched by a spring compressed 15 cm. The spring constant is 120 N/m. What speed does the cart have when it leaves the spring on a level track?

6. A 0.60 kg cart moving at 2.0 m/s runs into a 150 N/m spring bumper on a level frictionless track. How far does the spring compress before the cart stops?

7. A spring with $k = 80 \text{ N/m}$ is compressed 0.25 m. It launches a 0.50 kg cart up a frictionless ramp. How high does the cart rise before it stops?

8. A pole vaulter sprints down the runway at 9.0 m/s. If all of that kinetic energy became gravitational potential energy, how high could the vaulter raise their center of mass?

9. A 500 kg roller coaster car starts from rest at point A, 20 m above the ground. Ignoring friction, fill in the table for points B and C.

point	height (m)	U_g (J)	K (J)	speed (m/s)
A	20	?	0	0
B	5	?	?	?
C	12	?	?	?

10. A 70 kg skier starts from rest at the top of a 12 m tall hill. Ignoring friction, how fast is the skier moving at the bottom? Why doesn't the mass matter?

11. A 0.40 kg ball is moving at 3.0 m/s when it is 1.2 m above the floor. Ignoring air resistance, how fast is it moving when it reaches the floor?
12. From the top of a cliff, you throw one rock straight up at 10 m/s and an identical rock straight down at 10 m/s. Ignoring air resistance, which rock is moving faster when it hits the ground below?
13. A cart moves at 7.0 m/s at the bottom of a frictionless track. It reaches the top of a hill still moving at 3.0 m/s. How tall is the hill?
14. A 1.2 kg ball is thrown straight up at 5.0 m/s. A motion sensor shows that the ball stops rising after 0.51 s. Use kinematics to find how high it rises, then check your answer with conservation of energy.
15. A 900 kg car is moving at 18 km/h. If all of its kinetic energy could be stored in a giant spring bumper with $k = 20\,000\text{ N/m}$, how far would the spring compress?

16. A 0.20 kg cart starts at rest against a spring compressed 10 cm. The spring constant is 180 N/m. The cart leaves the spring and climbs a frictionless track. What is its speed after it rises 0.50 m?
17. A 0.30 kg block slides from rest down a frictionless track, dropping 1.4 m in height. At the bottom it runs into a 400 N/m spring. How far does the spring compress when the block first stops?
18. A 0.010 kg bullet is fired into a 2.0 kg wooden block hanging from strings. The bullet stays in the block, and the block swings up 0.32 m. How fast was the bullet moving?
19. Two equal-mass hockey pucks collide elastically on frictionless ice. Puck A moves right at 4.0 m/s, and puck B moves left at 1.0 m/s. What are their velocities after the collision?
20. A 2.0 kg cart moving right at 4.0 m/s collides elastically with a 6.0 kg cart at rest. After the collision, the 2.0 kg cart bounces left at 2.0 m/s. Use conservation of momentum to find the 6.0 kg cart's velocity, then check that kinetic energy is conserved.